1️⃣ Employ 12 (Most Relevant)

Use a multi-step procedure leading to a mathematical solution. employ

Reason

This is not a single-step calculation, but a **multi-step problem**, including:

1. Extracting roof height (sloped dimension) from the front view
2. Extracting length (6.00 m) from the side view
3. Determining the sloped length of the roof (possibly using the Pythagorean theorem)
4. Calculating the area of one rectangular section
5. Multiplying by 2 to obtain the total area

👉 This is a typical example of:

a multi-step modelling and calculation process

Therefore, this is the **core competency**.

2️⃣ Employ 6

Develop mathematical diagrams and extract mathematical information from them. employ

Reason

The problem provides:

- a front view (with height and width)
- a side view (with length)

Students must:

👉 extract key geometric information from the diagrams

For example:

- roof height
- half-width of the building
- roof length

This reflects:

the ability to extract mathematical information from diagrams

3️⃣ Employ 7

Manipulate numbers, graphical and statistical data and information. employ

Reason

Students need to process numerical data such as:

- length (6.00)
- height (1.00)
- width (2.50, etc.)

and perform operations such as:

- multiplication
- squaring and square roots

This reflects:

👉 **numerical and geometric data manipulation**

4️⃣ Employ 5

Understand and utilise constructs based on definitions, rules and formal systems. employ

Reason

Students need to apply:

- the area formula for rectangles: Area = length × width
- possibly the Pythagorean theorem (to determine sloped length)

This reflects:

👉 **application of geometric rules and formulas**

5️⃣ Employ 10

Make sense of, relate and use a variety of representations. employ

Reason

The problem involves multiple representations:

- a 3D model
- a front view
- a side view

Students must:

👉 integrate information across these representations

For example:

- front view → provides roof height
- side view → provides roof length

This reflects:

the ability to connect multiple representations

6️⃣ Employ 1 (Less Relevant)

Perform a simple calculation. employ

Reason

Although calculations are involved:

👉 individual calculations are not the main focus but serve the overall modelling process

Thus, its relevance is lower.

1️⃣ Employ 12（相关性最高）

Use a multi-step procedure leading to a mathematical solution. employ

体现原因

本题不是单一步骤计算，而是一个**多步骤问题**：

1. 从正视图提取屋顶高度（斜边信息）
2. 从侧视图提取长度（6.00 m）
3. 计算屋顶斜面长度（可能涉及勾股关系）
4. 计算一个矩形面积
5. ×2 得到总面积

👉 整个过程是典型的：

多步骤建模与计算过程

因此是本题最核心能力。

2️⃣ Employ 6

Develop mathematical diagrams and extract mathematical information from them. employ

体现原因

题目提供：

- 正视图（含高度、宽度）
- 侧视图（含长度）

学生需要：

👉 从图中提取关键几何信息

例如：

- 屋顶斜边高度
- 建筑宽度的一半
- 屋顶长度

这体现了：

从几何图形中提取信息的能力

3️⃣ Employ 7

Manipulate numbers, graphical and statistical data and information. employ

体现原因

学生需要对数据进行处理：

- 长度（6.00）
- 高度（1.00）
- 宽度（2.50 等）

并进行：

- 运算（乘法、平方、开方）

体现：

👉 **数值与几何数据操作能力**

4️⃣ Employ 5

Understand and utilise constructs based on definitions, rules and formal systems. employ

体现原因

需要用到：

- 矩形面积公式：Area = length × width
- 可能涉及勾股定理（斜边计算）

体现：

👉 **几何公式与数学规则的应用**

5️⃣ Employ 10

Make sense of, relate and use a variety of representations. employ

体现原因

题目涉及多种表示：

- 3D模型图
- 正视图
- 侧视图

学生需要：

👉 在不同表示之间建立联系

例如：

- 正视图 → 提供屋顶高度
- 侧视图 → 提供屋顶长度

体现：

多表示整合能力

6️⃣ Employ 1（次相关）

Perform a simple calculation. employ

体现原因

虽然包含计算，但：

👉 单个计算不是核心，而是服务于整体建模

因此相关性较低。